



UPES ACM & ACM-W

100 Days of Code

Day: 1

UPES ACM & ACM-W 100 Days of Code Platform

Problem 1

Find the Sum of Array Elements

Description

A problem that introduces array traversal techniques and teaches how to efficiently process every element while maintaining a running total of values present in an array.

Given an array of integers, the task is to find the sum of all elements in the array.

This problem appears frequently in interviews and real-world applications like analytics processing, financial calculations, reporting systems, and data aggregation tasks. The challenge involves traversing the array efficiently and accumulating the result.

This teaches:

- array traversal
- accumulation logic
- variable tracking
- optimization techniques

which are essential for efficient data processing and interview problem solving.

Input

An array of integers.

Your Task: Find the sum of all array elements using an optimized approach.

Output

Print the sum of all elements in the array.

Sample Input 1	Sample Output 1
Array = [12, 35, 1, 10, 34, 1]	93

Explanation

$12 + 35 + 1 + 10 + 34 + 1 = 93$

Sample Input 2	Sample Output 2
Array = [10, 5, 10]	25

Problem 2

Missing Number (LeetCode #268)

<https://leetcode.com/problems/missing-number/description/>

Description

A problem that teaches array traversal and logical reasoning techniques by identifying the missing value from a sequence of numbers.

Given an array containing n distinct numbers from the range $[0, n]$, find the number that is missing from the array.

This problem is commonly asked in interviews and helps build concepts like:

- array processing
- efficient traversal
- mathematical reasoning
- optimization techniques

which are important for solving data-processing problems efficiently.

Input

An array of n distinct integers in the range $[0, n]$.

Your Task: Find the missing number using an optimized approach.

Constraints: $0 \leq n \leq 10^4$

Output

Print the single missing number from the array.

Sample Input 1	Sample Output 1
Array = [3, 0, 1]	2

Explanation

$n = 3$, so numbers should be in the range $[0, 3]$. The array contains $\{0, 1, 3\}$, so 2 is missing.

Sample Input 2	Sample Output 2
Array = [0, 1]	2

Sample Input 3	Sample Output 3
Array = [9, 6, 4, 2, 3, 5, 7, 0, 1]	8